Bio-GO-SHIP DNA Collection Protocol

Version 1.0

Authors

Principal Investigators: Luke Thompson

NOAA AOML, USA

Adam Martiny

Department of Earth System Science, University of California, Irvine, USA

Version	Date Implemented	Description of Revisions
1.0	2023-11-27	Initial release by AOML

Overview

This protocol describes collection and filtration of marine eDNA samples using Sterivex cartridge filters for both CTD Niskin rosettes and underway flow-through systems. Water is collected with sterile pre-rinsed carboys and pumped through a Sterivex filter using a peristaltic pump. The recommended filtration volume for GO-SHIP samples is 8 L, which takes ~2 hours. However, the volume is reduced in high productivity regions to avoid filter clogging. Volume is to be recorded.
Videos of the protocol are here:
eDNA Sampling for Bio-GO-SHIP
Adding ZR beads to Sterivex filters (direct pour method)
Adding ZR beads to Sterivex filters (funnel pour method)

Supplies

Equipment:
Rinsed carboy with a defined volume (8 or 9 L)
Peristaltic pump with tubing (Cole-Parmer; cat# EW-96400-24)
Hose-barb adapter to connect filter inlet to tubing (Cole-Parmer; cat# EW-45505-19)
–80 or –20 °C freezer

Consumables:
Sterivex 0.22-µm cartridge filters (Millipore; cat# SVGPL10RC)
Filter inlet caps (VWR cat# MFLX45504-56)
Filter output caps (VWR cat# MFLX45508-64)
Pre-printed Sterivex filter labels
Pre-cleaned 60-mL syringe with male luer-lok outlet
Zymo ZR BashingBead Lysis Tubes (0.1 & 0.5 mm; cat# S6012-50)

Chemicals:
5% bleach (made from “100%” household bleach)
Deionized or Milli-Q water
DNA/RNA Shield (Zymo; CAT# R1100-250)

Collection and filtration protocol

- Prepare the Sterivex filters. Add 1 tube of Zymo ZR BashingBeads to the Sterivex filter (see videos above). Label filter with a pre-printed sticker.

- Wearing gloves, collect water from the flow-through system (or Niskin bottles) into a carboy. Rinse the carboy 3 times with sample water before filling.
    - Note: You may be using a spigoted carboy or a carboy without a spigot.

- Connect tubing to carboy and turn on the pump and run ~100 mL of seawater to prime the tubing. Run the pump at 100–150 rpm. Wear gloves throughout filtration and handling.
    - If using a spigoted carboy, you connect tubing to spigot. If you are using a non-spigoted carboy, you will use a large pipette tip attached to tubing that will be inserted into the carboy.

- Pause the pump and discard water.

- Attach the filter input to the hose-barb that is already attached to the tubing outflow. Filter should screw on tightly. Avoid handling filter input or output.

- Run the pump and filter seawater until ~8 L has been filtered

- On the log sheet, record the date, time, latitude, longitude, volume filtered, and any notes about the sample. (Every day, enter this information into an Excel spreadsheet to maintain a digital copy in addition to the paper copy.)

- Pause pump and open pump head values to release pressure. Unscrew and remove Sterivex filters from the hose-barb adapters. Gently remove any excess seawater from the filters using a sterilized 60-mL syringe (pre-cleaned with 2% bleach).

- Cap the bottom end (outlet) of the Sterivex capsule with a female luer-lok cap.
    - Note: If Sterivex has a slip-end, use critoseal to seal this end before moving to the next step.
    - Additional option with slip-end: Wrap with small piece of parafilm and cap with a red silicon cap

- Using a P1000 (1000-µL pipette), add preservative: 1000 µL DNA/RNA Shield into Sterivex.

- Cap the top end (inlet) of the capsule with a male luer-lok plug.

- Freeze at ≥ –80 °C until extraction. Freezing at -20˚C is acceptable for shorter periods but not long-term storage.

- Return the bottle of lysis buffer or DNA/RNA Shield to a refrigerator to reduce contamination.

- Run ~1 L of 5% bleach through the lines to clean pump tubing. Repeat with ~1 L of deionized water.

- Rinse the collection carboy 3 times with DI water after sampling.
    - Note: If you are using spigoted carboys, you will
        1. Finish filtering sample and rinse the carboy with tap water
        2. Fill carboy with 1 L of 5% bleach and run bleach through tubing
        4. Rinse carboy with DI water to remove trace bleach
        5. Fill carboy with 1 L of DI water and run DI through tubing

Sampling Scheme 1 (no bio casts):
Underway sampling at 6 am, 12 pm and 8 pm local time. Duplicate sample collected every 3 days.

Sampling Scheme 2 (with one daily bio cast):
Underway sampling at 6 am and 8 pm local time.
CTD rosette sampling at 12 pm at four depths (0, 100, 200, and 1000 m)
Duplicate sample taken every 2 days.